



Integrated Antiship Missile Project

Solid-propellant sizing, point-mass guidance, DATCOM/6DOF dynamics and autopilot design

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Contents

1	Project Overview	1
2	Preliminary Sizing	2
2.1	Main Assumptions	2
2.2	Sizing Results	2
3	Point-Mass Guidance Prototype	4
3.1	Base Model and Changes	4
3.2	Guidance Cases	5
4	DATCOM Aerodynamics and 6DOF Model	8
4.1	Coefficient Generation	8
4.2	CG and Inertia	9
4.3	6DOF Results	10
5	Attitude and Acceleration Autopilots	13
5.1	Autopilot Architecture	13
5.2	Gain Schedule	13
5.3	Root Locus and Step Responses	16
5.4	Engagement Check With Autopilot Constraints	18
6	Integrated Interpretation	21
6.1	Course Code Reuse	21
6.2	Main Engineering Decisions	21
6.3	Limitations	21
6.4	Final Summary	21
7	Reproducibility Procedure	23
7.1	Required Tools	23
7.2	Execution Order	23
7.3	Data Consistency Checks	23



1 Project Overview

This report consolidates the four partial assignments into one antiship missile design path. The starting point is an Exocet/MM40-class equivalent missile constrained to solid propulsion. From that initial sizing, the model is carried through guidance, aerodynamic data generation, 6DOF preparation and autopilot design.

The work was developed in stages, with each stage passing data or design constraints to the next one:

Table 1: Role of each partial report in the integrated project.

Stage	Main model	Role in the integrated project
Report 1	Preliminary sizing and 1D propulsion simulation	Defines mass, diameter, length, booster, sustainer, solid propellant and range capability.
Report 2	Horizontal point-mass guidance simulation	First guidance version: pre-programmed trajectory, seeker logic and terminal switching.
Report 3	DATCOM coefficients and 6DOF model	Converts geometry into aerodynamic data, CG/inertia and a higher-fidelity dynamic model.
Report 4	Root-locus autopilot design	Uses the 6DOF/DATCOM data to design attitude and acceleration autopilots.

In this sequence, Report 2 works as the first guidance prototype. Reports 3 and 4 then replace part of that simplified representation with DATCOM aerodynamics, mass properties and control design. The project flow is therefore:

Sizing $\rightarrow$ Point-mass guidance $\rightarrow$ DATCOM/6DOF dynamics $\rightarrow$ Autopilots

The resulting model has 0.35 m diameter, 5.8 m length, 855 kg launch mass and simulated range above 70 km. It also includes horizontal-plane terminal guidance, a DATCOM aerodynamic database, a section-based CG/inertia model and autopilots satisfying the specified $5g$ and 0.5 Hz requirements.



2 Preliminary Sizing

The first stage sized a solid-propellant antiship missile equivalent to the Exocet/MM40 class. The professor's sizing script was used as reference: atmospheric properties were computed at low altitude, drag was estimated at cruise, the booster accelerated the missile to cruise Mach number, and the sustainer was sized to balance cruise drag for the required range.

2.1 Main Assumptions

- Cruise speed: Mach 0.9;
- Required range: 70 km;
- Reference diameter: 0.35 m;
- Solid booster plus solid sustainer;
- Sustainer thrust approximately equal to cruise drag.

2.2 Sizing Results

Table 2: Final preliminary missile dimensions and masses.

Parameter	Value
Diameter	0.35 m
Total length	5.80 m
Launch mass	855 kg
Empty mass	508.20 kg
Total propellant mass	346.80 kg
Propellant fraction	40.6%
Cruise speed	306.23 m/s
Simulated range	71.09 km

Table 3: Solid propulsion sizing.

Parameter	Booster	Sustainer
Burn time	4.00 s	228.59 s
Mean thrust	65.46 kN	2.319 kN
Propellant mass	121.44 kg	225.36 kg
Grain length	1.047 m	2.286 m
Specific impulse	220 s	240 s



Exocet-class solid propulsion simulation

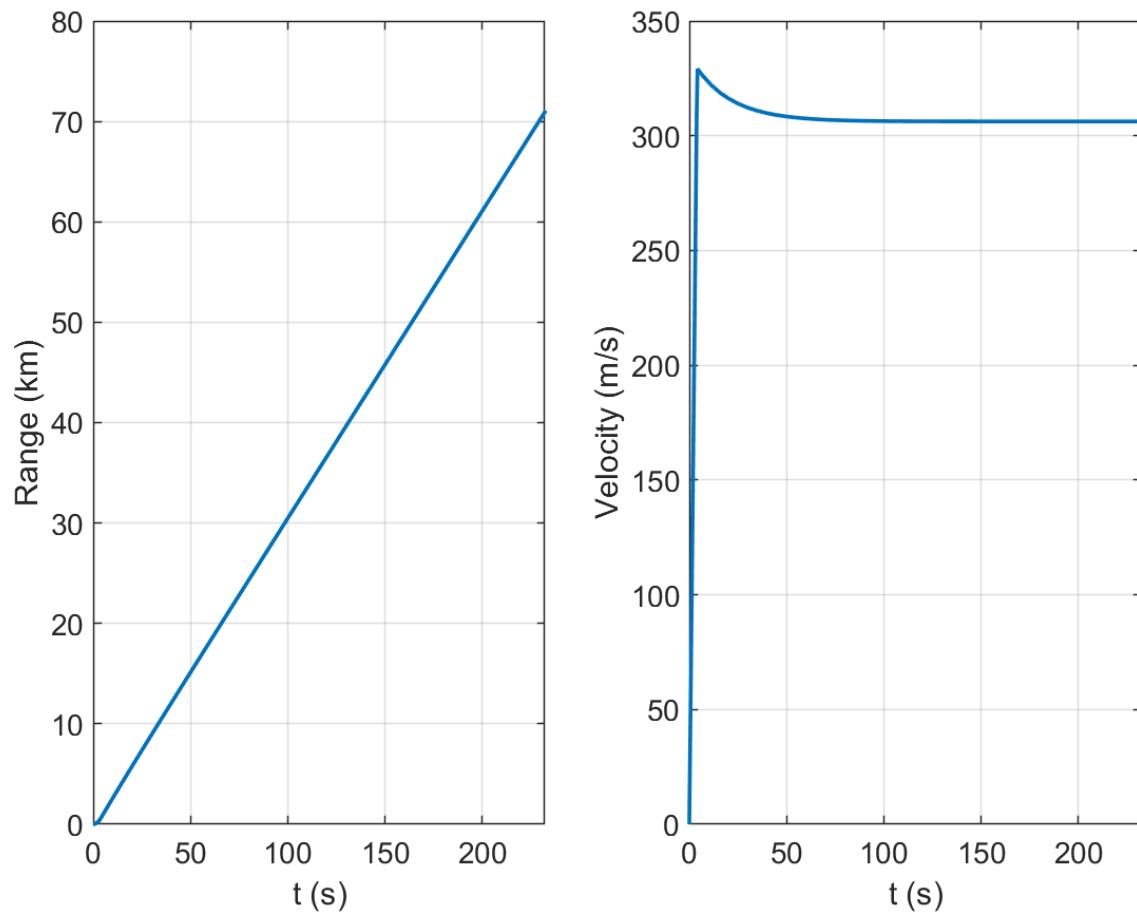


Figure 1: Range and velocity from the preliminary propulsion simulation.

The sizing simulation reached 71.09 km, above the 70 km requirement. This result became the basis for later mass, propulsion and geometry definitions.



3 Point-Mass Guidance Prototype

The second stage used the MANSUP point-mass model to test two horizontal engagement geometries. At this level, the model is mainly useful for guidance logic: pre-programmed flight, seeker limits and the switch to terminal homing.

3.1 Base Model and Changes

The course Simulink model remained as the base, with edits concentrated in the control subsystem:

- the X - Z plane was interpreted as the horizontal plane;
- Z was treated as horizontal lateral displacement;
- altitude 10 m was used only for atmosphere calculation;
- a pre-programmed turn law was added for off-boresight launch;
- terminal guidance was allowed only when the target was within the required distance and seeker cone.

The pre-programmed command was based on the angular error between the missile heading and the line of sight:

$$G_{pre} = K_{pre} [\text{atan2}(\Delta z, \Delta x) - \gamma].$$

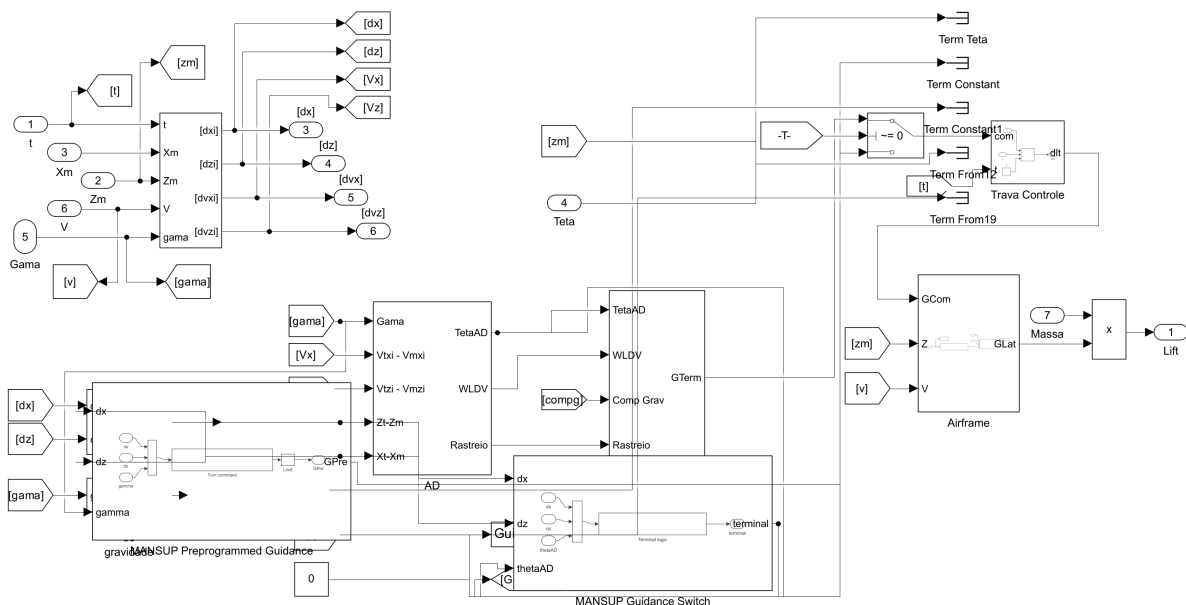


Figure 2: Modified control subsystem in the point-mass Simulink model.

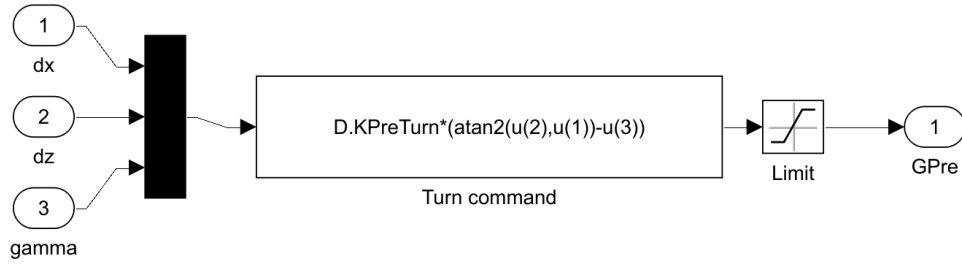


Figure 3: Pre-programmed guidance block used to generate the horizontal turn command.

3.2 Guidance Cases

Table 4: Point-mass simulation cases.

Case	Initial condition	Terminal logic
1	Target at 65 km, launch bearing 0°	Terminal guidance starts at 20 km from the target.
2	Target at 15 km, launch bearing 90°	Seeker is on from launch, but terminal mode waits until target enters the 40° cone.

Table 5: Point-mass simulation results.

Result	Case 1	Case 2
Terminal guidance start	146.35 s	13.95 s
Range at terminal start	19.997 km	12.680 km
Flight time	214.90 s	58.95 s
Final range	500.0 m	549.0 m
Final speed	284.4 m/s	305.0 m/s

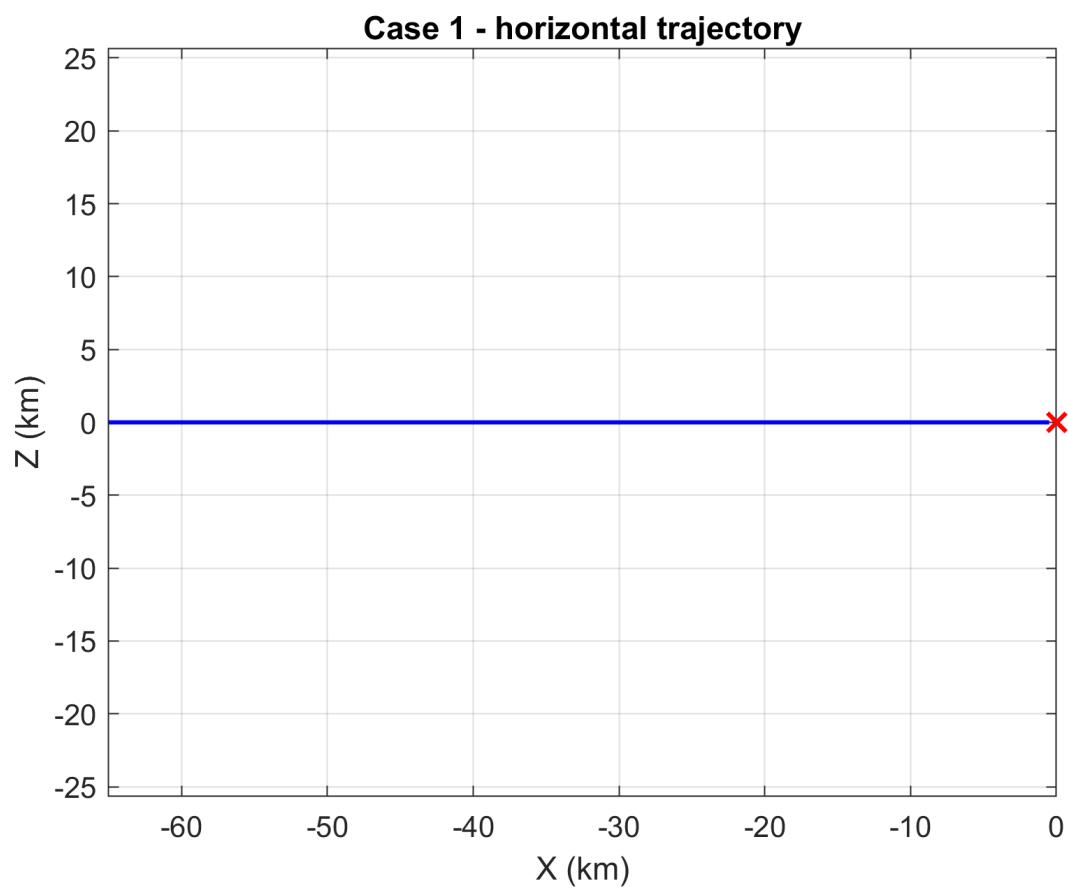


Figure 4: Case 1 horizontal trajectory.

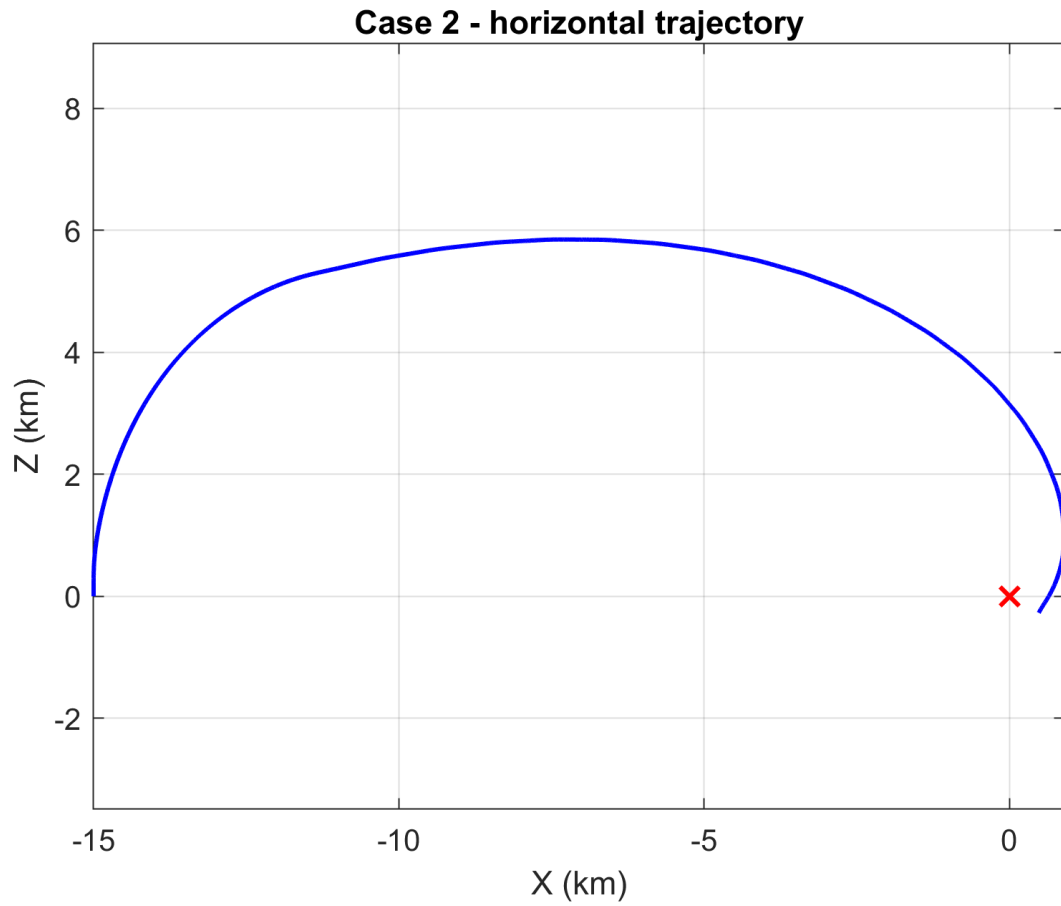


Figure 5: Case 2 horizontal trajectory with pre-programmed turn before terminal guidance.

The switching logic uses two conditions: the seeker must be active and the target must be inside the allowed field of view. This is the important point in Case 2, because the seeker starts active while the target is still outside the 40° cone.



4 DATCOM Aerodynamics and 6DOF Model

The third stage increased model fidelity. The equivalent antiship missile geometry was converted into a DATCOM coefficient set, and a section-based mass model was used to estimate CG and inertia for 6DOF simulation.

4.1 Coefficient Generation

The workflow preserved the course processing chain:

`for005.dat` $\rightarrow$ `Misdat/RunDatcom2` $\rightarrow$ `coeficientes.dat` $\rightarrow$ `Coef2Mat` $\rightarrow$ `M_aed.mat`.

The geometry writer was adapted to the Exocet-equivalent missile. The reference point $X_{ref} = 2.700$ m was later reused in the 6DOF model as the common moment reference.

Table 6: Main DATCOM setup values.

Parameter	Value
Moment reference	2.700 m
Diameter	0.350 m
Nose length	0.700 m
Center-body length	5.090 m
Control deflection sweep	-20° to 20°

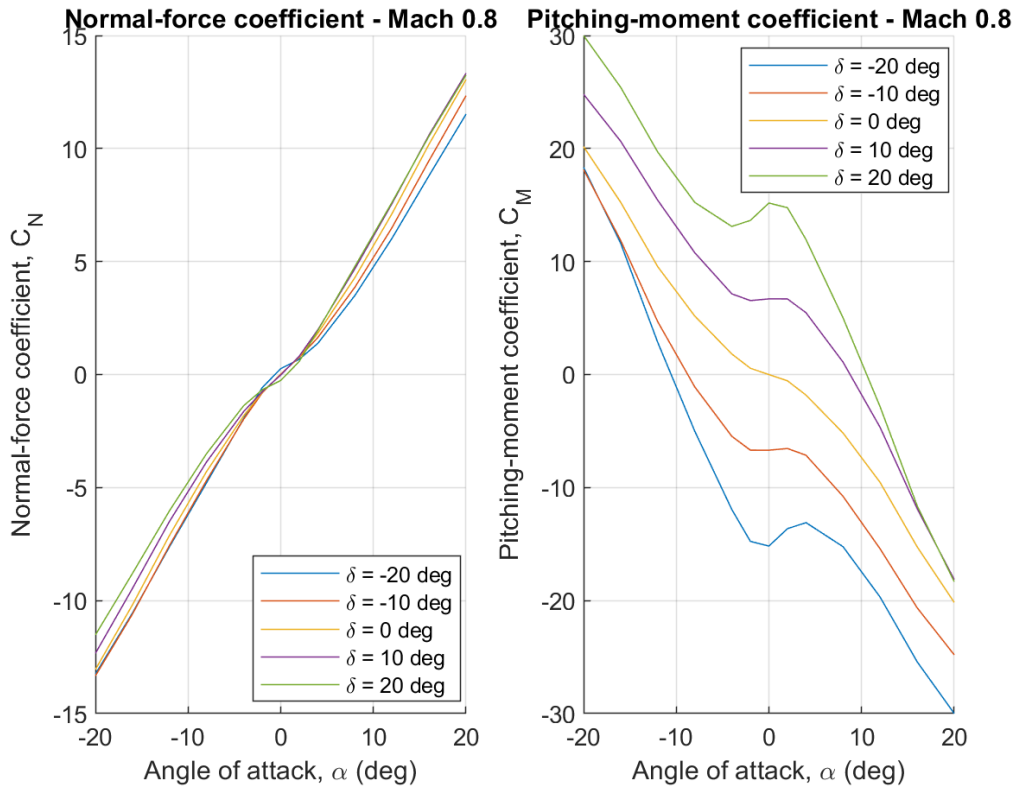


Figure 6: Generated normal-force and pitching-moment coefficients.

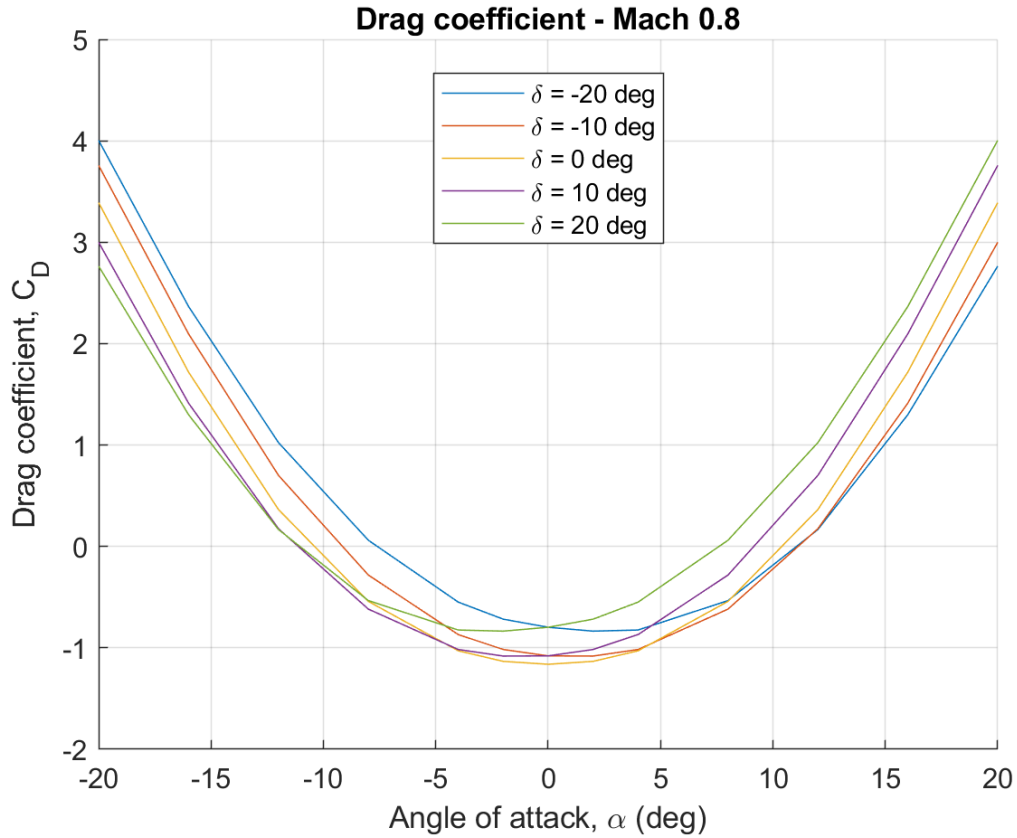


Figure 7: Generated drag coefficient.

At Mach 0.9, $\alpha = 0^\circ$ and $\delta = 0^\circ$, the coefficient set gave $C_{M\alpha} = -15.36 \text{ rad}^{-1}$ and $C_{N\alpha} = 22.70 \text{ rad}^{-1}$. The DATCOM value $X_{CP} = -0.677$ is dimensionless in calibers relative to the moment reference; converting it gives $X_{CP} = 2.937 \text{ m}$ from the nose, about 0.30 calibers aft of the estimated initial CG. This is consistent with static stability.

4.2 CG and Inertia

The missile was divided into control/seeker section, warhead, booster, sustainer and nozzle. The same moment reference used in DATCOM was inserted in `DadosCGInercia.m`:

```
D.CRM = [2.700 0 0]';
```

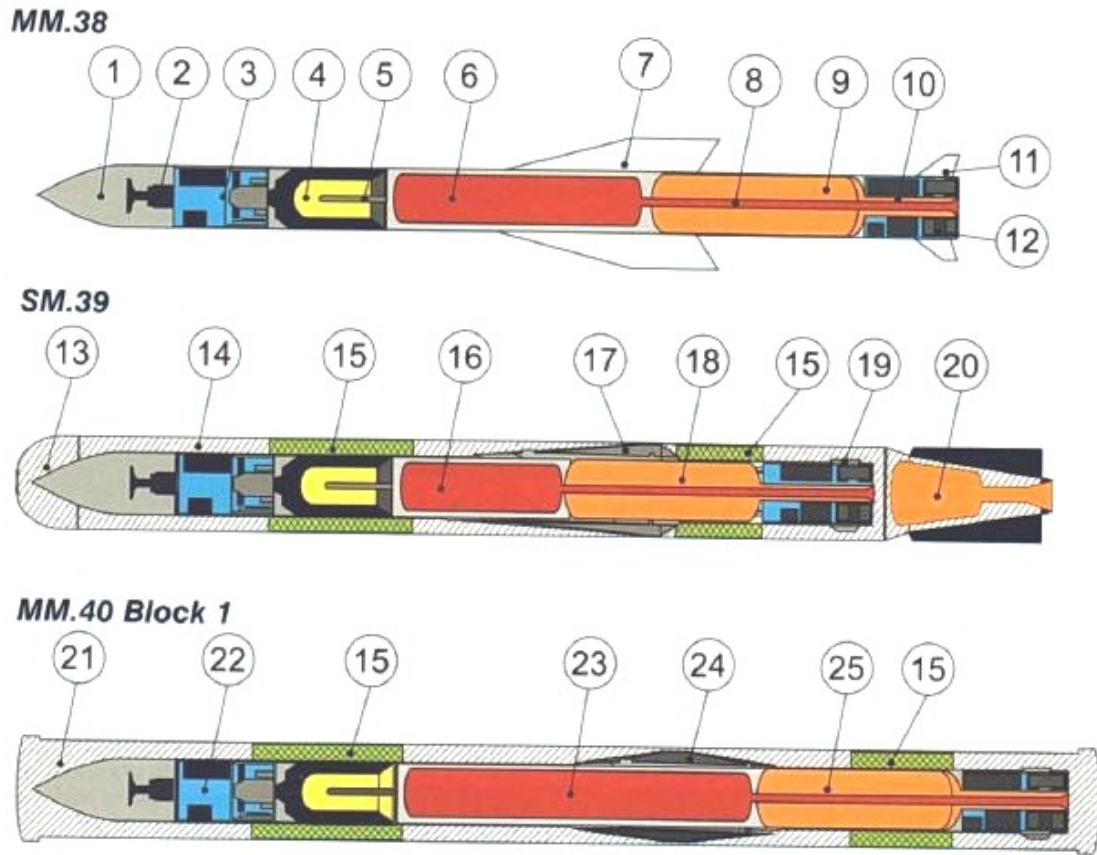


Figure 8: Internal subsystem reference used for mass distribution.

Table 7: Initial and final mass properties.

Parameter	Initial	Final after burnout
Mass	874.9 kg	365.0 kg
X_{CG}	2.833 m	1.856 m
I_{xx}	13.40 kg.m ²	5.59 kg.m ²
$I_{yy} = I_{zz}$	2155.40 kg.m ²	863.30 kg.m ²

4.3 6DOF Results

The 6DOF test was a short dynamic check during the initial flight phase, focused on the relationship between control deflection and angle of attack.

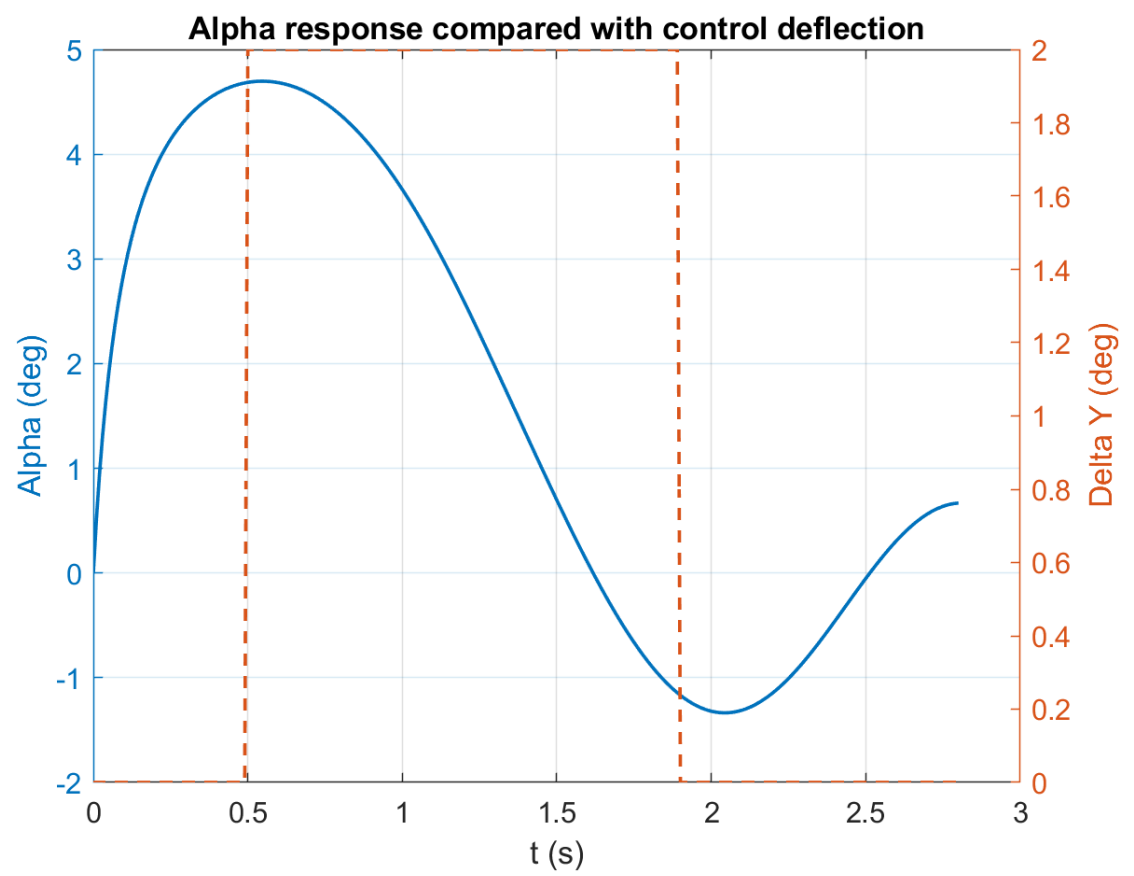


Figure 9: 6DOF comparison between angle of attack and control deflection.



Velocity and accelerations

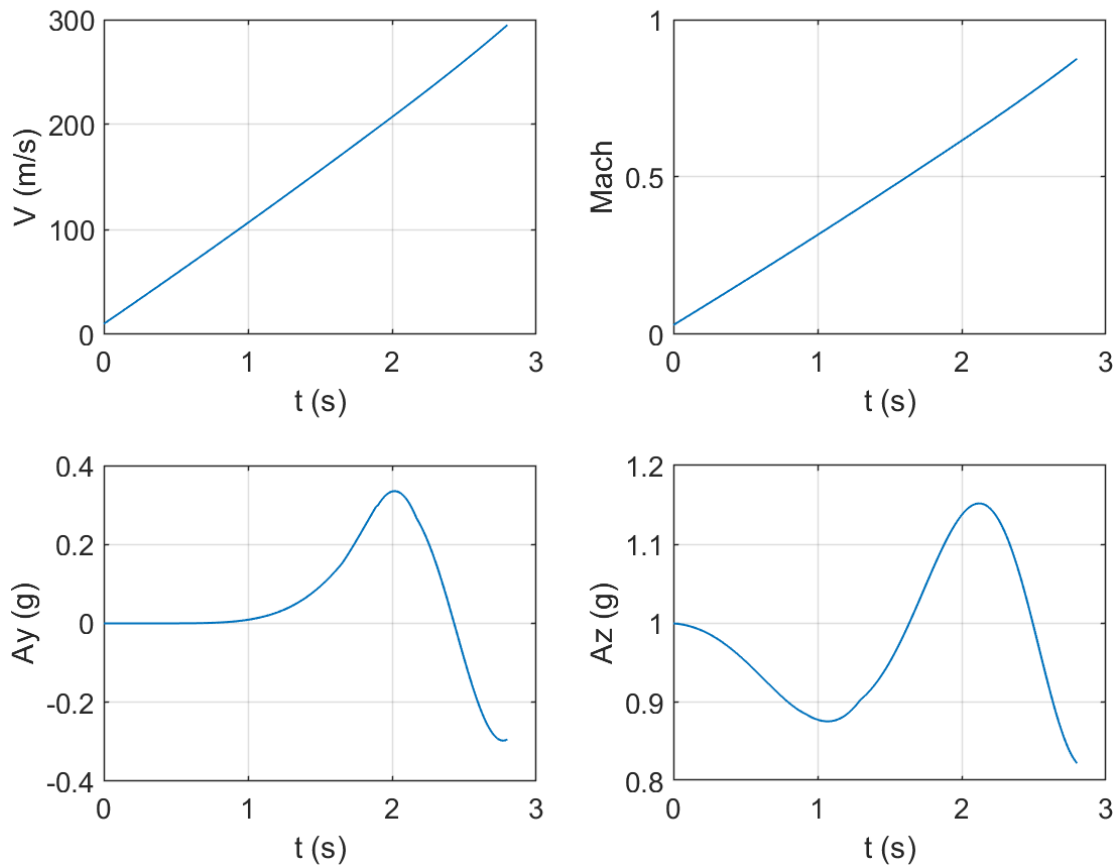


Figure 10: Velocity and acceleration in the 6DOF test.

The short simulation served as a consistency check before autopilot design: aerodynamic database, mass properties and 6DOF equations had to respond coherently in the same model.



5 Attitude and Acceleration Autopilots

The fourth stage used the course folder 6DOF - CURSO/03-autopiloto. The professor's root-locus script was copied and adapted for the antiship missile. The same support files remained in the workflow:

- FTransDin.m;
- calcula_coef.m;
- DadosControle.m;
- Autopiloto.slx.

The missile data, CG/inertia and aerodynamic matrix were replaced by the antiship missile data from the previous stage.

5.1 Autopilot Architecture

The attitude autopilot uses an inner q -feedback loop and an outer attitude loop:

$$\theta_c \rightarrow K_\theta \rightarrow (K_q, \text{ actuator}, q/\delta) \rightarrow \frac{1}{s} \rightarrow \theta.$$

The acceleration autopilot uses the same inner q -feedback loop and an outer acceleration loop with an integrator:

$$A_{z,c} \rightarrow \frac{K_a}{s} \rightarrow (K_q, \text{ actuator}, q/\delta) \rightarrow A_z/\delta \rightarrow A_z.$$

5.2 Gain Schedule

Table 8: Computed autopilot gain schedule.

t (s)	X_{CG} (m)	K_q attitude	K_θ	K_q accel.	K_a
0.000	2.8325	0.6000	9.3333	0.6000	0.0250
1.725	2.8934	0.6667	10.6667	0.6667	0.0250
3.450	2.9647	0.7333	12.0000	0.7333	0.0250
115.350	2.6012	0.3333	8.0000	0.3333	0.0250

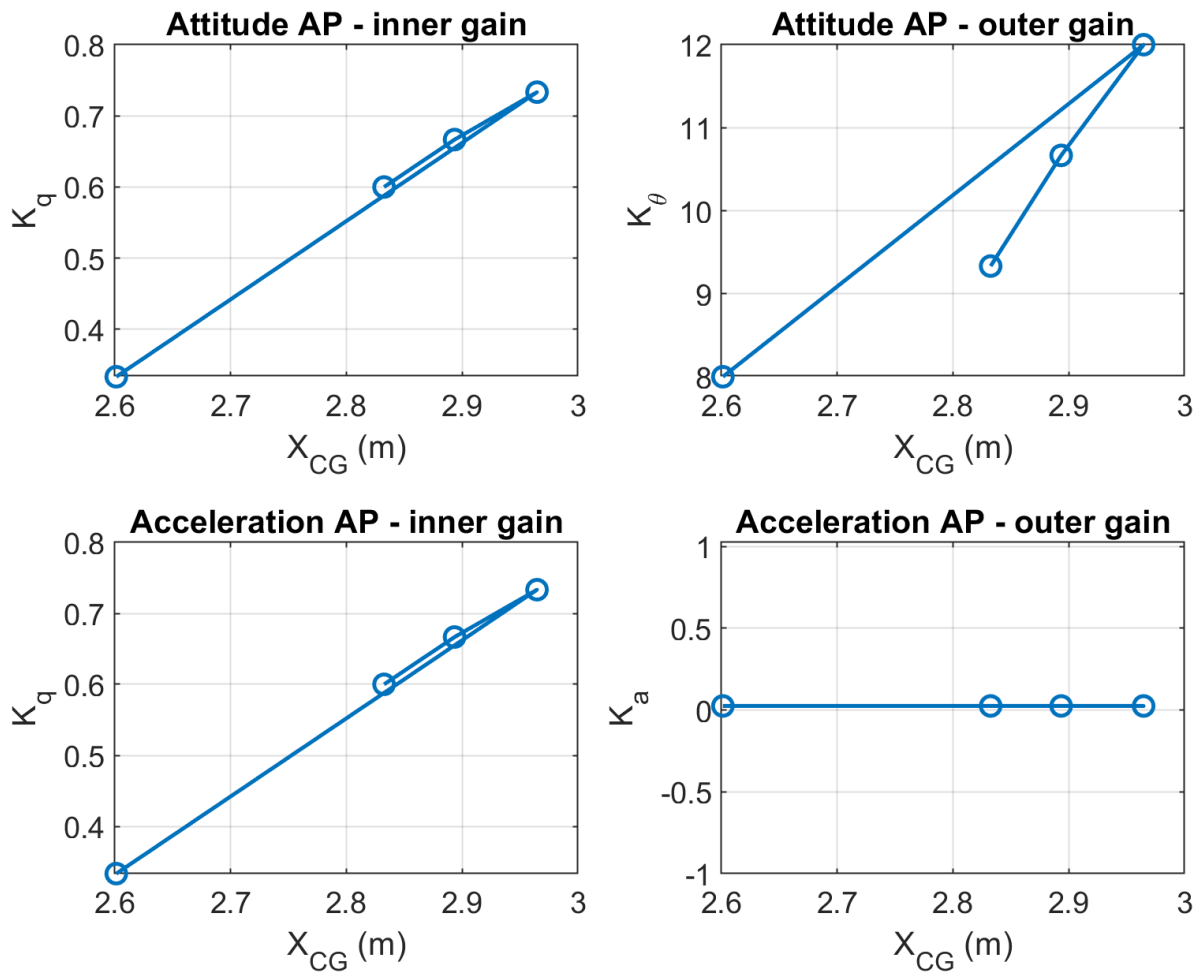


Figure 11: Internal and external gains as a function of CG.

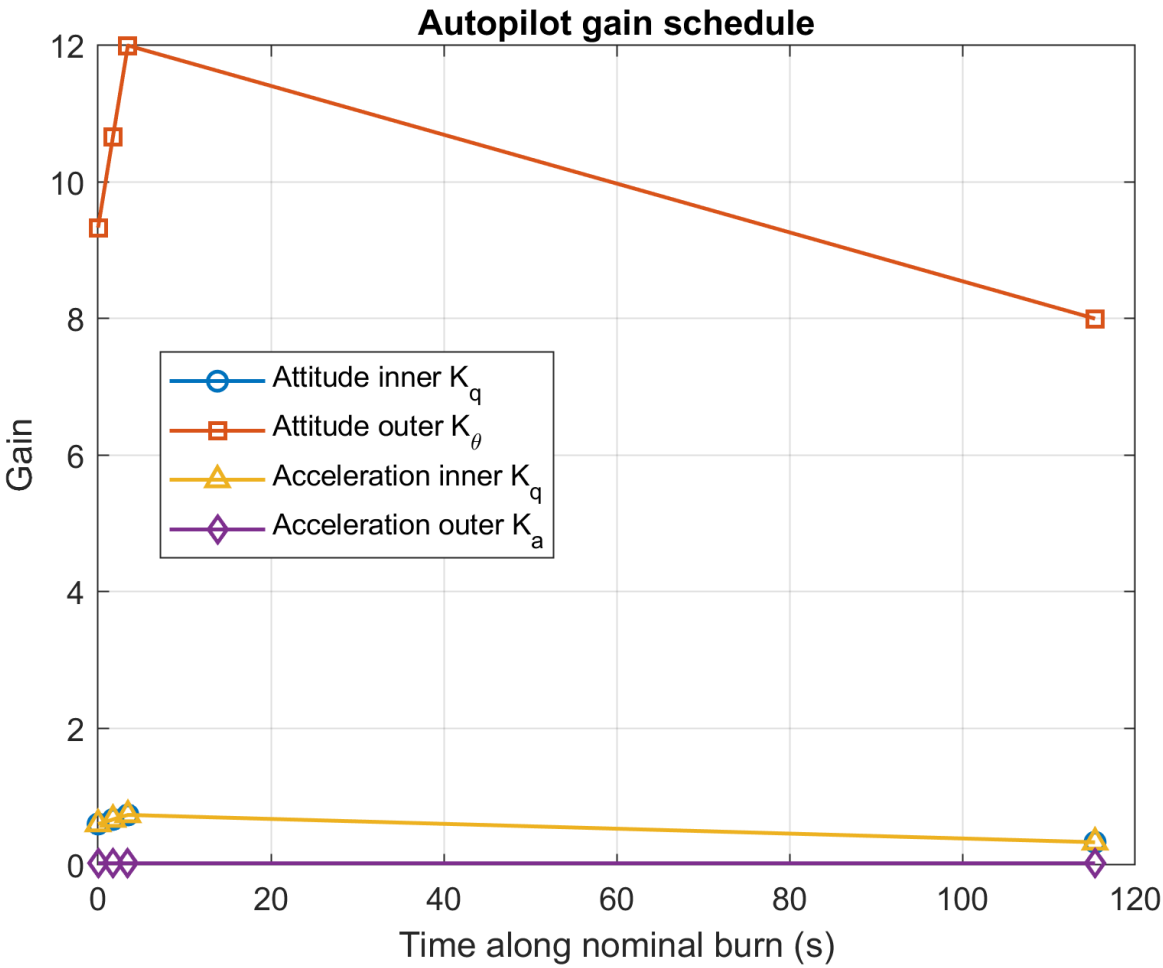


Figure 12: Autopilot gain schedule along representative burn points.



5.3 Root Locus and Step Responses

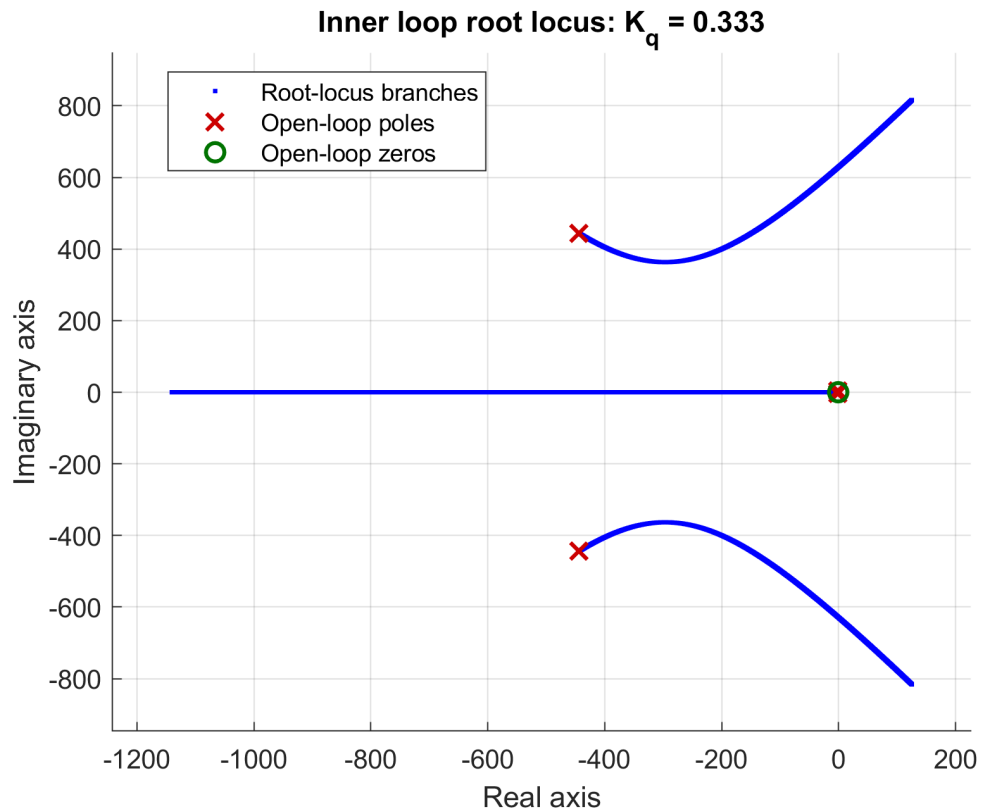


Figure 13: Root locus of the common internal q -feedback loop.

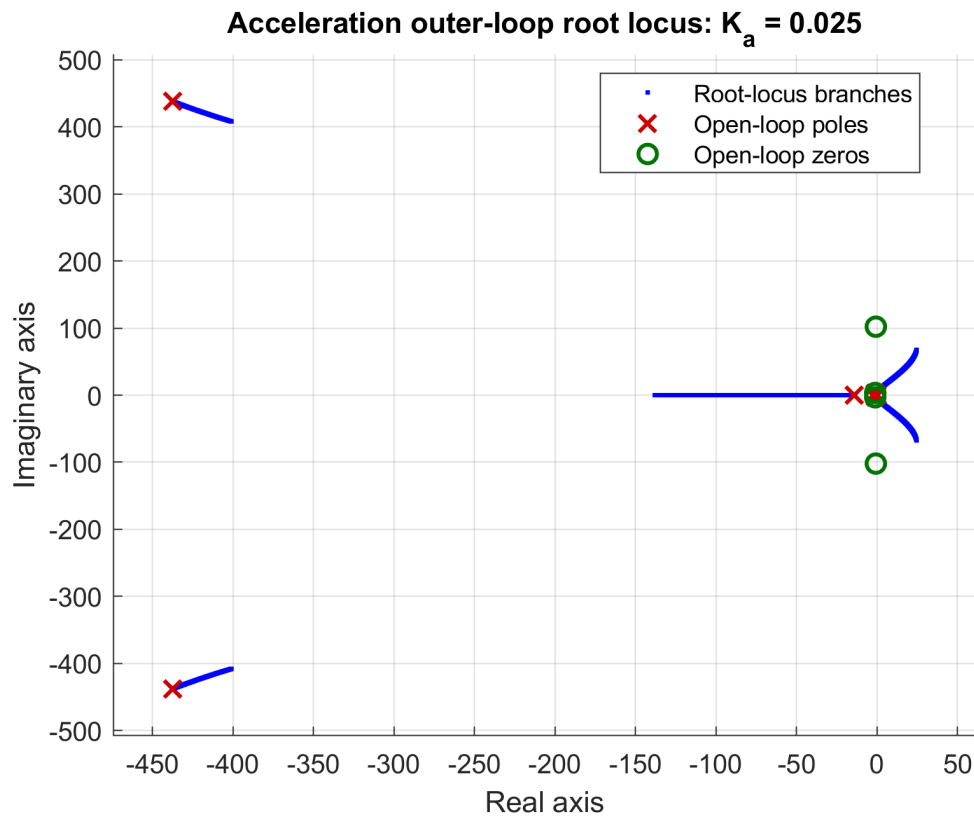


Figure 14: Root locus of the external acceleration loop.

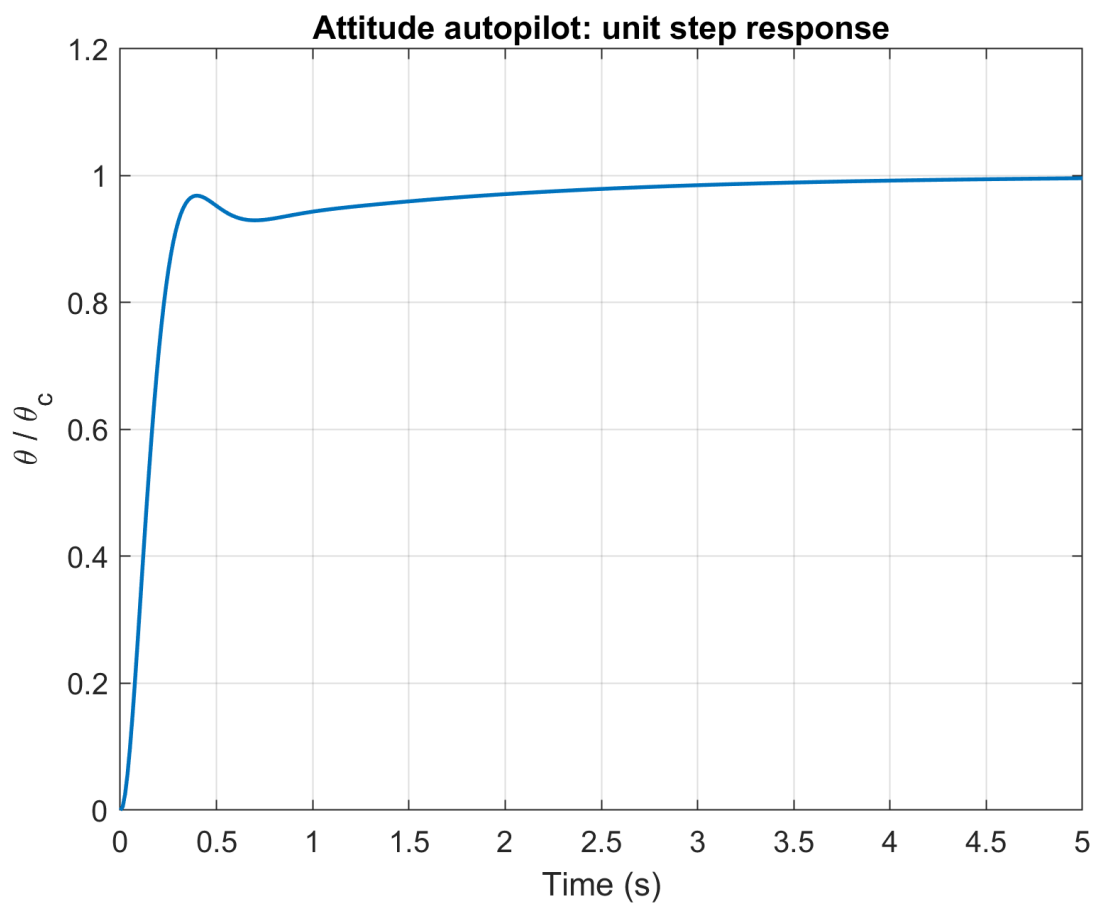


Figure 15: Attitude autopilot unit-step response.

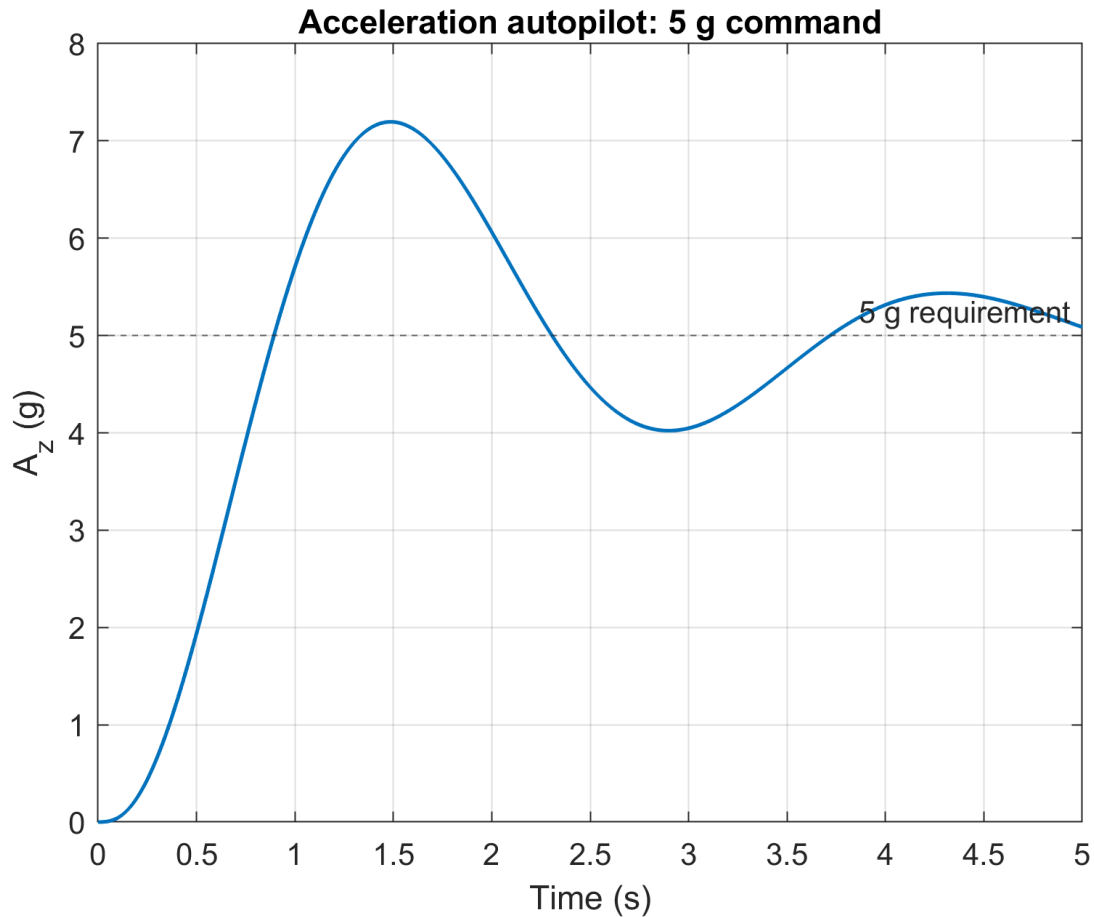


Figure 16: Acceleration autopilot response to a 5g command.

Table 9: Nominal requirement check.

Requirement	Result	Limit	Status
Lateral acceleration capability	13.52g	$\geq 5g$	Satisfied
Attitude autopilot bandwidth	1.652 Hz	≥ 0.5 Hz	Satisfied
Acceleration autopilot bandwidth	0.537 Hz	≥ 0.5 Hz	Satisfied

The acceleration loop was the limiting case. The initial root-locus gain was too aggressive for the nominal model, so the external gain was adjusted to $K_a = 0.025$. This kept the loop stable while maintaining bandwidth above 0.5 Hz.

5.4 Engagement Check With Autopilot Constraints

The final check connects the autopilot limits to the mission geometry. A horizontal point-mass engagement was run with Mach 0.9 at 10 m, 5g lateral acceleration saturation, seeker cone of 40° , and an airframe response of $w_n = 0.5$ Hz. Terminal guidance used proportional navigation. Case 1 switched to terminal guidance at 20 km; Case 2 first used a pre-programmed turn until the target entered the seeker cone.



Table 10: Engagement results using R4 constraints.

Case	Minimum range	Time	Status
65 km, bearing 0°	48.93 m	212.10 s	Satisfied
15 km, bearing 90°	47.06 m	54.43 s	Satisfied

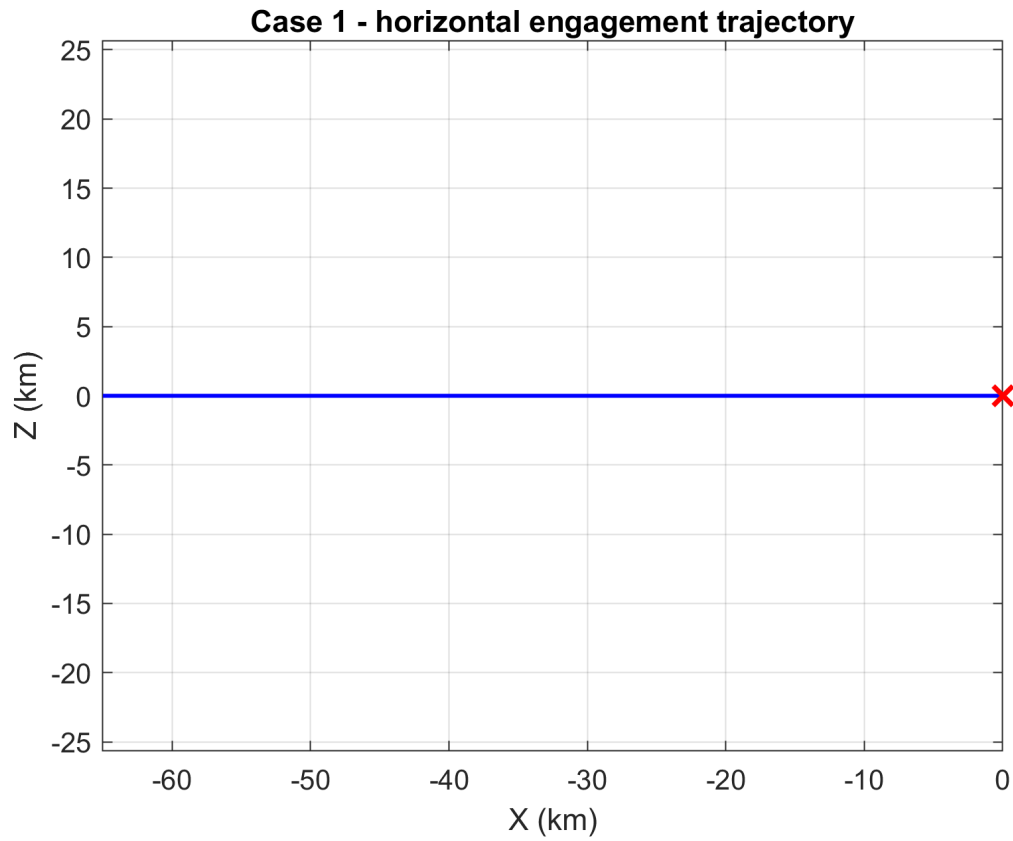


Figure 17: Case 1 trajectory with R4 constraints.

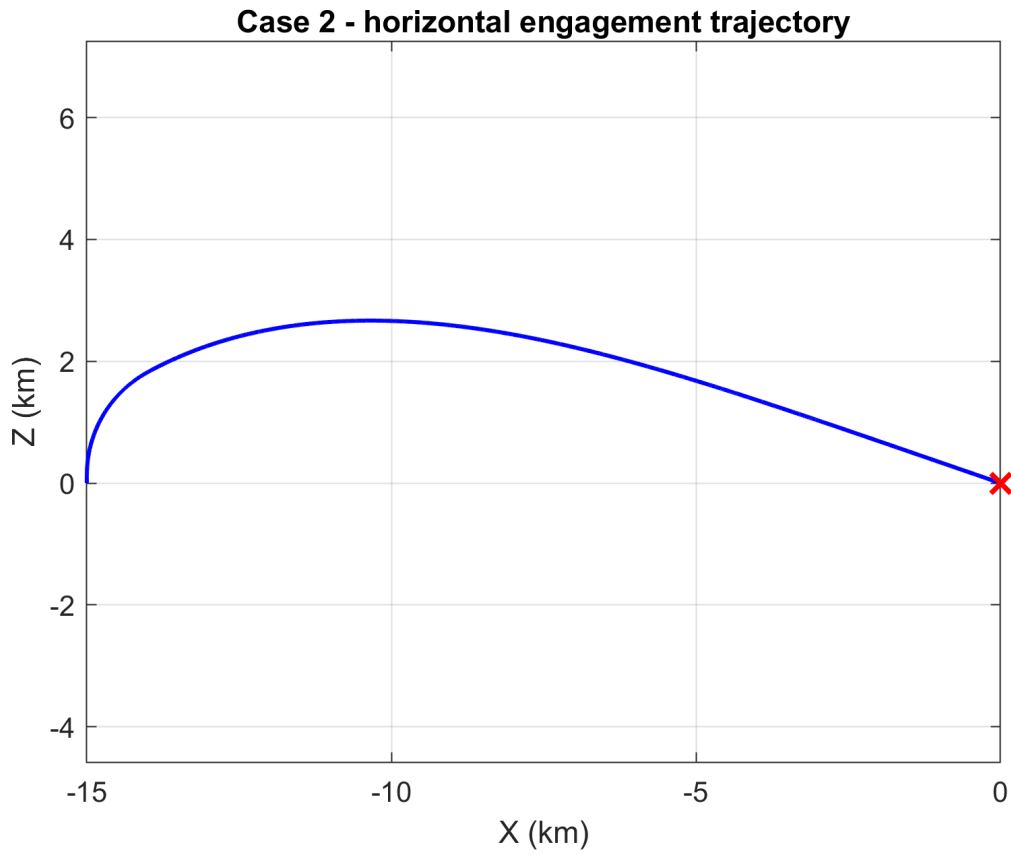


Figure 18: Case 2 trajectory with R4 constraints.

Both engagement cases pass the 50 m miss-distance criterion. The second case is the sizing case for maneuvering because it reaches the $5g$ lateral acceleration limit during the initial turn.



6 Integrated Interpretation

The four assignments fit as successive layers of the same preliminary missile design. The first layer fixes the vehicle scale and propulsion balance. The second checks whether the missile can be guided in the horizontal plane under the seeker limits. The third produces the aerodynamic and mass-property data needed for a 6DOF model. The fourth uses those data to size attitude and acceleration autopilots.

6.1 Course Code Reuse

- Preliminary propulsion: the professor's sizing structure and `SimX` verification were retained.
- Point-mass guidance: the MANSUP Simulink model was reused, with edits focused on pre-programmed guidance and terminal switching.
- DATCOM: the processing chain with `DATCOMreader.m` and `Coef2Mat.m` was retained; the geometry input was rebuilt for the selected missile.
- 6DOF and autopilot: the course dynamic model and root-locus method were reused with the new missile data, CG/inertia and aerodynamic matrix.

6.2 Main Engineering Decisions

The decisions that most affected the final results were:

- using solid propulsion only, requiring a long-burning solid sustainer;
- separating seeker activation from terminal-guidance authorization;
- using the same moment reference in DATCOM and 6DOF;
- correcting the DATCOM X_{CP} interpretation as a dimensionless value in calibers;
- tuning the acceleration autopilot gain around the root-locus result to satisfy both stability and bandwidth.

6.3 Limitations

The remaining gap is the integration between guidance and 6DOF control. The point-mass guidance model and the 6DOF/autopilot model were developed in separate course frameworks; a complete end-to-end Simulink mission remains the next integration step. Propulsion, structural masses and aerodynamic geometry are preliminary engineering estimates, and the autopilot was checked at representative points rather than over a full flight envelope. Within that scope, the data flow is consistent: geometry, mass properties, aerodynamic coefficients and control gains refer to the same missile configuration.

6.4 Final Summary

The final design package contains:

- simulated range above 70 km;
- horizontal guidance logic for straight and off-boresight launches;



- DATCOM aerodynamic database for the equivalent geometry;
- section-based CG and inertia estimates;
- 6DOF dynamic checks;
- attitude and acceleration autopilots meeting $5g$ and 0.5 Hz requirements.



7 Reproducibility Procedure

The project can be reproduced by running each stage in the order below. The sequence keeps the same structure used in the course material: first the missile is sized, then guidance is checked in the point-mass model, then aerodynamic and mass-property data are generated, and finally the autopilots are designed.

7.1 Required Tools

- MATLAB with Simulink;
- Control System Toolbox;
- the course point-mass and 6DOF folders;
- the course DATCOM workflow with `RunDatcom2` and `Coef2Mat.m`.

7.2 Execution Order

Table 11: Minimum execution sequence to reproduce the delivered results.

Stage	File or action	Expected output
Sizing	run the solid-propulsion sizing script and <code>SimX</code>	missile mass, length, booster/sustainer data, range and velocity plot
Point-mass guidance	run the MANSUP horizontal cases script and the adapted Simulink model	trajectories, seeker/terminal switching times and final miss distances for both cases
DATCOM geometry	generate <code>for005.dat</code> from the adapted geometry script	DATCOM input file with the same moment reference used later in 6DOF
DATCOM processing	run <code>RunDatcom2</code> , then <code>Coef2Mat</code>	<code>coeficientes.dat</code> and <code>M_aed.mat</code>
Coefficient plots	run <code>PlotCmCN.m</code> , <code>PlotCD.m</code> and <code>PlotCoef.m</code>	aerodynamic coefficient figures
6DOF data	edit/run <code>DadosMissil.m</code> , <code>DadosCGInercia.m</code> and <code>VooUnico.m</code>	CG/inertia values and dynamic response plots
Autopilots	run <code>ProjRLocus_antiship.m</code> using the course autopilot files	gain curves, root-locus plots, step responses and requirement check

7.3 Data Consistency Checks

Three checks are important before comparing results. First, the diameter, length, mass and propulsion values from the sizing stage must be the same values used to build the DATCOM geometry and the CG/inertia model. Second, the DATCOM moment reference must match the 6DOF reference:

$$D.CRM = [2.700 \ 0 \ 0]';$$

Third, DATCOM XCP must be treated as a dimensionless value in calibers. With $X_{ref} = 2.700$ m, $D = 0.350$ m and $XCP = -0.677$,

$$X_{CP,abs} = 2.700 - (-0.677)(0.350) = 2.937 \text{ m.}$$



These checks keep the aerodynamic moments, CG position and autopilot design referenced to the same physical missile.